

line to cease the ring. At the same time, the high output of IC5 (pin 3) drives relay RL1 to enable the bell for a preset time period.

The decoder cum display section is built around DTMF IC 8870 (IC1), IC 7447 (IC2), common-anode 7-segment display LT5542 (DI51) and some discrete components. The main IC of this section is MT8870. It is widely used in telephones and a variety of other applications. The DTMF (dual-tone multi-frequency) tone pair generated by pressing the telephone button is converted into binary values by MT8870. The binary values are fed to driver IC

7447 to display corresponding number on the seven-segment display.

Working of the circuit is simple. Connect the assembled circuit to dedicated extension for the office boy. Connect 5V to the circuit using 5V power adaptor. Now when you dial this dedicated extension number, the ring will cease in a few seconds. After the ring ceases, press a digit on the telephone to indicate your location (say, 5). This number will be displayed on the seven-segment display and the bell will sound for a preset time period to alert the office boy. After providing the assistance to this room, the office

boy can wait for the next call.

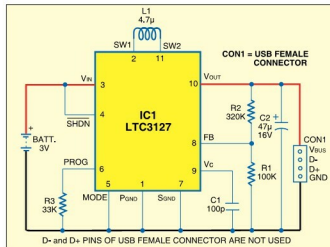
If you want to use the telephone as a regular intercom, simply connect switch S2 to position 1. This will disable the circuit. Now you can dial any extension number through the telephone. If you want to use the call bell again, just turn switch S2 to position 2.

Assemble the circuit on a general-purpose PCB and enclose in a suitable cabinet. Fix the connectors to connect the telephone line and AC supply at the rear side of the cabinet. Mount the 7-segment display, buzzer and switch S2 on front side of the cabinet. ●

USB Charger for Mobile Phones

DILEEP CHACKO

All of us worry about our mobile phone draining out of charge while travelling. In such a situation, this SMPS-based USB charger powered by just two AA-size batteries will come handy.



The advantage of SMPS is high efficiency and low cost. Of course, most mobile phones can also be charged through the standard USB port available in desktop and laptop computers.

A unit load is 100 mA in USB 2.0, and 150 mA in USB 3.0. An individual

device may draw a maximum of 5 units load (500 mA) from USB 2.0 and 6 units (900 mA) from USB 3.0.

This USB charger for mobile phones operates off a 2.4V or 3V battery. It is designed around an average input-current-controlled buck-boost DC-DC converter LTC3127 (IC1) and some discrete components. The circuit converts 3V or 2.4V input into 5V output.

Pins SW1 (pin 2) and SW2 (pin 11) are used to connect the inductor. SHDN (pin 4) shuts down the output of LTC3127 when low. Pin 5 is used to change the mode of operation between PWM and burst modes. If Mode pin is low the mode will be PWM; when it is high the mode will change to burst mode. PROG (pin 6) is used to set the average input current limit threshold. Vc (pin 9) is used to connect the compensation components, which impart stability

to this regulator.

The values of R1 and R2 decide the output voltage level as follows:

$$V_{OUT} = 1.195 \times 1 + R2R1$$

You can select appropriate values

Pin Details of USB Connector

Pin	Name	Description
1	V _{BUS}	+5V
2	D-	Data-
3	D+	Data+
4	GND	Ground

of R1 and R2 to get 5V output (V_{OUT}).

Most mobile phones have a USB male connector for charging, so a female connector is required for the charger side. Assemble the circuit on a small PCB and enclose in a suitable cabinet. Use a USB female connector and solder the connections as shown in the figure. The table gives the pin details of a USB connector. For powering the circuit, use an AA-size battery pack having either two 1.5V alkaline batteries or two 1.2V (Ni-Cd or Li-ion) batteries. ●